



JAVA BACKEND ENGINEERING / GEOSPATIAL SYSTEMS

Alireza Nobakht

Java Developer | Spring Boot Backend Engineer

[E alireza.nobakht2004@gmail.com](mailto:alireza.nobakht2004@gmail.com) [M \(+98\) 910-977-7936](tel:+989109777936) [L Tehran, Iran](#)
[GH github.com/alirezanobakht2004](https://github.com/alirezanobakht2004) [IN linkedin.com/in/alireza-nobakht](https://linkedin.com/in/alireza-nobakht) [W Portfolio](#)

PROFESSIONAL PROFILE

Java Developer with **2+ years of software-engineering experience** building Spring Boot APIs, Hibernate-backed data services, geospatial platforms, and Android field applications. Experienced with Java, Spring Framework, RESTful services, PostgreSQL/PostGIS, clean architecture, database optimization, offline synchronization, and connected-device workflows. Combines backend engineering with practical GPS/IMU and embedded-systems experience to deliver reliable software for real-world field operations.

01 TECHNICAL SKILLS

JAVA & SPRING

Java, Spring Boot, Spring Framework, Spring MVC, REST APIs, dependency injection, OOP, async and concurrent programming concepts

PERSISTENCE & DATA

JPA / Hibernate ORM, PostgreSQL, PostGIS, SQL Server, database schemas, spatial indexes, query optimization, Redis caching

ARCHITECTURE & APIS

Clean Architecture,DDD, SOLID, design patterns, layered applications, DTO mapping, validation, standardized error handling, OpenAPI concepts

MOBILE & WEB

Java Android SDK, Kotlin / Android, Thymeleaf, React, JavaScript, HTML, CSS, Leaflet, OpenLayers, offline-first synchronization

DELIVERY & QUALITY

Git, Docker, CI/CD pipelines, GitHub Actions, Linux fundamentals, unit and integration testing concepts, monitoring, production debugging

CONNECTED ENGINEERING

STM32, C/C++, Qt, C#, UART, SPI, I2C, GPS / IMU, serial protocols, multithreading, real-time processing, encrypted transfer

02 PROFESSIONAL EXPERIENCE

Software Developer & Embedded Systems Engineer

Feb 2024 - Present

Hiromap (Hirongar Pars Company) | hiromap.com | Tehran, Iran

- Developed a **Java Spring Boot geospatial platform** with PostgreSQL/PostGIS for storing, processing, and querying survey coordinates, maps, and field measurements.
- Designed **RESTful APIs** for data upload, retrieval, project workflows, and map rendering, with Hibernate ORM and spatial indexing for scalable access to large datasets.
- Integrated **React with Leaflet and OpenLayers** to deliver interactive map visualization, survey-point management, and topographic-layer overlays.
- Built a **Java Android field-survey application** with GPS tracking, offline caching, validation, and synchronization to a Spring Boot server in low-connectivity environments.
- Created a **Spring Boot + Thymeleaf administration dashboard** for survey projects, users, and field reports; real-time validation and automated coordinate correction reduced manual-entry errors by **40%**.
- Produced **C++/Qt and C# diagnostic tools** for 115200-baud serial streams with zero observed data loss and 40% lower memory use, supporting connected GPS/IMU workflows.

Technical Instructor & Educational Consultant

Aug 2022 - Jun 2023

Ghalamchi Educational Institute | Tehran, Iran

- Mentored students during a **10-month engagement** in programming fundamentals, algorithms, problem solving, and software-development practices.
- Prepared **structured lessons and hands-on exercises** that strengthened technical communication, collaboration, and systematic debugging skills.

Projects, Education & Development

Spring Boot backends, geospatial data, Android field software, and connected engineering.

JAVA DEVELOPER
SPRING BOOT
ENGINEER

Geospatial Data Management System

Location-based survey-data platform

Java Spring Boot Hibernate ORM PostgreSQL / PostGIS REST React Leaflet

Apr 2025

- Built a **Spring Boot and PostgreSQL/PostGIS platform** to store, analyze, and visualize survey points, coordinates, field measurements, and topographic data.
- Implemented **RESTful services** for data upload, retrieval, processing, and map rendering, backed by Hibernate ORM and optimized spatial queries.
- Connected **React with Leaflet/OpenLayers** for interactive layers, survey-point management, and efficient access to large geospatial datasets.

Field Survey Data Collection App

Offline-capable mobile surveying workflow

Java / Android SDK Spring Boot REST APIs GPS Offline Storage Thymeleaf

Jul 2024

- Developed a **Java Android application** for recording geotagged coordinates, notes, images, and field observations directly on site.
- Implemented **GPS tracking, local caching, and secure REST synchronization** with a central Spring Boot service to support low-connectivity environments.
- Added **real-time validation and coordinate-correction workflows**, reducing manual input errors by **40%** and improving field-data reliability.

Desktop Diagnostics & Cross-Platform Transfer

Real-time tooling and device integration

- Produced **C++/Qt** and **C#** applications for streaming, filtering, visualizing, recording, and analyzing serial data.
- Supported **CSV, JSON, and binary exports** with asynchronous buffering and configurable communication parameters.
- Engineered encrypted **Bluetooth/Wi-Fi file transfer** with progress, resume, retry, and chunking for files larger than 100 MB.

115200 baud processing 99.5% transfer success 35% faster transfer Qt C# Kotlin / Android Encryption

2024 - 2025

Surveying Rover Firmware

Navigation and real-time data collection

- Led **STM32 firmware development** for a 100 Hz control loop with PID motor control, GPS/IMU fusion, Kalman filtering, and power management.
- Established **UART and SPI telemetry** with fault recovery and error correction for difficult RF conditions.
- Achieved **+/-2 cm accuracy, 99.9% transmission reliability** over 500 m, and 8+ hours of runtime.

STM32 C/C++ GPS / IMU PID Kalman Filter

Jul 2024

EDUCATION

B.Sc. Computer Engineering

K.N. Toosi University of Technology
Sep 2022 - Present

Expected graduation: 2026. Teaching Assistant for core programming courses. Study areas include algorithms, databases, software engineering, operating systems, networks, embedded systems, and OOP.

PROFESSIONAL FOCUS

Java & Spring Ecosystem

Project-based development with Spring Boot, Hibernate ORM, REST APIs, PostgreSQL/PostGIS, Android, and clean architecture.

Ongoing study: scalable backend design, database performance, software architecture, and cloud fundamentals.

LANGUAGES & PRACTICES

Persian - Native

English - Intermediate / Professional
Technical documentation, independent work, cross-functional collaboration, code review, production debugging, and clear stakeholder communication.